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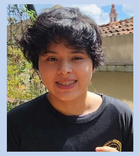
We learned to design our own hardware using AI and we validated it ourselves



RoboCupJunior Soccer · Open League · Salta, Argentina · National champions 2025 → RoboCup 2026, Incheon

ABSTRACT

What we're most proud of: we learned to design our own hardware using AI as a tool — and we validated it ourselves. We're students who didn't know how to design PCBs. For the world championship we rebuilt two robots with an AI-assisted workflow we call VIBE and learned to design our two new boards from scratch: TOP (perception) and DOWN (floor/line).

The robot is a 3-board Teensy architecture + comms module, each a specialist: TOP perceives and publishes a 31-byte WorldSnapshot; CENTRAL (reused 2025-champion Zircon) decides and drives 3 omni wheels; DOWN senses the floor. The striker has no kicker — it pushes by inertia.

The method we contribute is verifying firmware on the PC without the board: pure C++ modules tested offline — 858 tests / 61 suites / 0 failures. This poster documents the method and design so another junior team can replicate it.

OUR JOURNEY

Dec 2025 — national champions with a MUCH more basic robot: up top a single camera (no ToF or ultrasonic), at the bottom only 3 light sensors, all on the Zircon board... and we still won Argentina's 1st national RoboCupJunior Soccer competition.

2026 — the leap: same champion brain (Zircon), much more perception. Top: 2 cameras + IMU + 4 ToF + ultrasonic. Bottom: ring of 32 sensors + 2 OTOS. Redesigned to 3 boards adding perception (TOP) and floor (DOWN) — continuity, not replacement.

May–Jun 2026: bring-up of the 3 physical boards, ~30 documented bench sessions, host test suite growing 246 → 262 → 324 → 354 → 403 → 470 → 545 → 658 → 834 → 858 / 0 failures.

Jun–Jul 2026: Incheon. Team's declared strategy: invest in learning, play honest matches and capture data.

Team: María & Elías (competitors) + Enzo (head coach) travel to Incheon · Gustavo (director/ coordinator). AI was the tool — we're responsible.

Fig.1 — Our journey: 2025 national-champion robot vs 2026 robot. Original team photo, CC BY 4.0.

OUR PROUD FEATURE

we learned to design hardware with AI ("VIBE" methodology)

We didn't know how to design electronic boards. The most important thing about this project isn't a sensor: it's that we learned to design them with AI as a tool. How we did it, concretely:

- VIBE PCB Design — the two new boards (TOP and DOWN) were designed almost entirely with AI (Claude Code via MCP to EasyEDA). It wasn't "we already knew": it was learning HOW to design PCBs with AI.
- Honest trial-and-error: we also tried commanding Flux with AI, but it did NOT work. We tell it because part of learning is what didn't work.
- VIBE 3D Design — motor-mount redesign in Fusion 360 via AI, now finished, to bring the motors further in. The full 3D design is used as the poster's anchor figure (Fig.5, exploded view of the 3 sections).
- VIBE Coding — C++ firmware with the safety net of hundreds of host tests, 0 failures (the guarantee: we validate on the PC before trusting; it's not copy-paste from the AI).

METHOD & DESIGN

Why 3 boards?

Each board processes where the sensor is and decisions are made in the center (standard mobile-robotics rule). This cuts UART traffic, targets each MCU <30% CPU (design goal), and lets us swap one board without touching the others.

2 modules, one clean data interface: an UPPER MODULE = perception + comms + sensor fusion (knowing where everything is — ball, goals, obstacles — fused into the WorldSnapshot) and a LOWER MODULE = drive train + decision brain (motors, drivers and game logic; the DOWN floor board sends its pre-processed line/ odometry data up). Each module is improved and tested separately → faster iteration.

TOP (Teensy 4.0) — "I see the world": 2 OpenMV N6 cameras · BNO055 IMU · 4 VL53L7CX ToF · HC-SR04 · GPIO referee

CENTRAL (Teensy 4.1, Zircon) — "I decide": tactical FSM · omni-3 kinematics · 3 PIDs · 3 motors

DOWN (Teensy 4.0) — "I touch the floor": 32 line sensors (4× CD4051 mux) · 2 OTOS

COMM (ESP32-C6) — RCJ referee: 3.3 V GPIO level to TOP

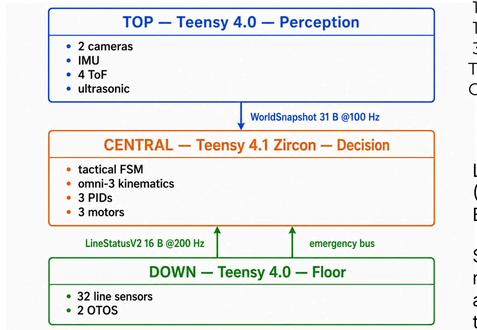


Fig.2 — Data flow between the 3 boards. Original team diagram, CC BY 4.0.

TOP
-2 cameras · IMU · 4 ToF
-1× HC-SR04 ultrasonic → frontal obstacle
- 2× OpenMV N6 camera (STM32N6 + NPU) → ball + goals
-2× BNO055 IMU → heading (yaw)
- 4× VL53L7CX ToF (reads 4×4 = 16 raw zones) → 2D localization

CENTRAL — Zircon · FSM · 3 motors

DOWN
-2× OTOS → pose/velocity from floor slip
-32× line ring (phototransistor) → field edge braking

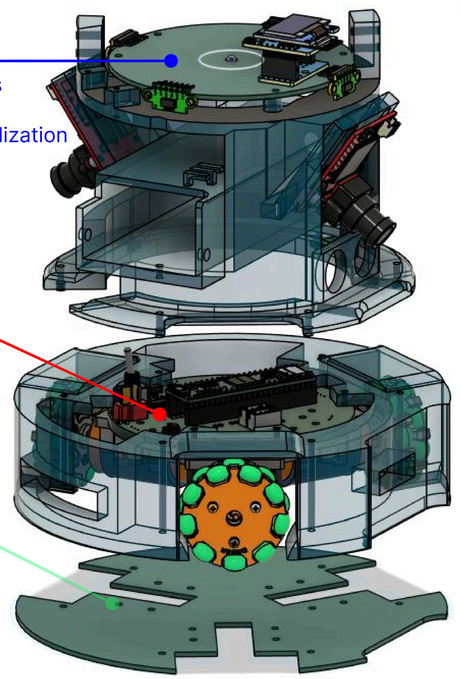


Fig.5 — The robot at a glance: exploded 3D render showing the 3 specialist boards + housings (crystal housings for visibility). Original team render, Fusion 360, CC BY 4.0.

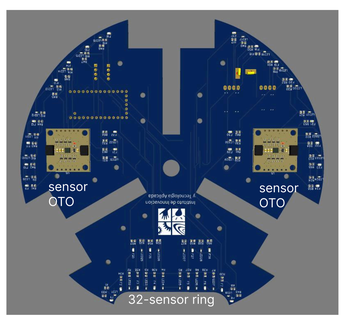


Fig.7 — 3D view (EasyEDA) of the DOWN board showing the line-sensor ring. Original team design, CC BY 4.0.

How much does it cost, and how long did it take?

2× OpenMV N6 camera — USD 165 ea (the most expensive)
2× Teensy 4.0 (TOP/DOWN) — USD 23.80 ea
1× Teensy 4.1 / Zircon (CENTRAL) — reused (2025 champion)
2× BNO055 IMU — ~USD 35 ea
4× VL53L7CX ToF — USD 19.95 ea
2× SparkFun OTOS — USD 84.95 ea
1× Zircon PCB (reused) — kit USD 529
1× LiPo 2S 6800 mAh — USD 42.99
3× DC motor + 3 omni wheels
TOTAL ≈ USD 1,168 new · ≈ USD 887 reusing CENTRAL

Development time — dated milestones:
Project kickoff — 2026-02-21
DOWN bring-up (32 sensors + 2 OTOS) — 2026-05-24
TOP bring-up (4 ToF, BNO) — 2026-05-25→31
Referee homologated (robot moves 1st time) — 2026-06-02/03
Full robot demo — 2026-06-11
TOP→CENTRAL validated on bench (66 Hz, 0 CRC) — 2026-06-14
Total effort ≈ 4 months (Feb–Jun 2026); intensive redesign ≈ 8 weeks on the 2025 champion base.

What does it sense, and with what code?

Languages: C++17 (firmware, packed structs with static_assert size checks) + MicroPython (OpenMV N6 vision, LAB color detection).
Build: PlatformIO (80+ [env:] environments) + an offline g++ runner for tests.

Sensing innovation
raw ToF zones (2026-06-14): before, the firmware averaged each ToF's 16 zones into ONE distance and threw away the rest. Now telemetry exposes each sensor's raw 4×4 grid — we see WHICH part of the wall is close, not just "how close"

DATA, RESULTS & DISCUSSION

Each row is a real iteration: we measured → evaluated → changed the source at a single point → re-verified. Maturity: BENCH-VALIDATED (on the real board) · HOST-VERIFIED (g++ tests / golden / compiles) · DIAGNOSED (cause found, bench fix pending).

#1 — 4 ToF collide on I²C (all born at 0×29; PCB didn't route XSHUT). After bodge + power-cycle: 4 sensors enumerate at 0×2A–0×2D → unlocks 2D localization. → BENCH-VALIDATED
#4 — Referee didn't reach the Teensy (TOP waited for a UART frame COMM never sends). Fix: read GPIO pins 5/6 with pulldown, fail-safe to STOP. → BENCH-VALIDATED — moved the robot for the 1st time
#8 — CENTRAL blind to the line (decoded the old 5 B contract, rejected 100% of real 16 B frames). Fix: migrate to LineStatusV2 (16 B). → HOST-VERIFIED
#9 — Ghost ball (fusion averaged 2 cameras → nonexistent midpoint). Fix: per-camera telemetry → decide WITH DATA which to disable. → HOST-VERIFIED
#15 — Tried commanding Flux with AI — didn't work. Switched to EasyEDA via MCP. We tell it because honest trial-and-error is part of learning.

Discussion: major changes #1, #4, #8 and #9 unlocked or protected entire capabilities (2D localization, referee homologation, edge braking, ghost-ball detection). We deliberately distinguish bench-validated from host-verified.

TEST METHODS (repeatable by any team)

M1 — Host-native verification (no board). Logic lives in pure C++ modules, compiled with g++ and run offline. Result: 858 tests / 61 suites / 0 failures (measured 2026-06-14). Live figure — re-measured each session. Needs a g++ in PATH (we use Webots') — without a compiler the script reports 0 tests, not failures.

M2 — Mandatory power-cycle on I²C bring-up. VL53L7CX and OTOS addresses persist with 3V3; a reset isn't enough. Protocol: flash → cut and restore power (10 s) → open monitor.

M3 — Odometry on textured surface. Move a controlled 300 mm and compare; tolerance 8% (result: 6.5%).

M5 — TOP→CENTRAL link bench-validated (2026-06-14). WorldSnapshot arrives at 66 Hz with 0 CRC and 0 seqGap, fully decoded. Proof the 3 boards really talk, not just in simulation.

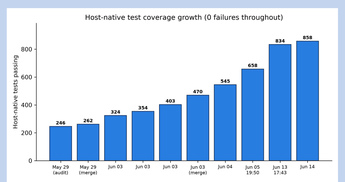


Fig.8 — Host test growth: 246 → 858 tests. Original team diagram, CC BY 4.0.

OPEN SOURCE (all MIT): Full firmware for the 3 boards (C++17) + vision (MicroPython) + hundreds of host tests, 0 failures · Complete EasyEDA projects for TOP & DOWN (schematic + PCB + Gerbers + BOM) · CENTRAL = reused Zircon

GitHub

github.com/IITA-Proyectos/open-soccer-robocup-team2026 · Instagram: @robocup_soccer.arg26

Instagram

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